

Mostafa Jafarian Abyaneh

College Park, MD | mjafaria@umd.edu | [Google Scholar](#) | [mostafa-aby.github.io](https://github.com/mostafa-aby)

SUMMARY

Ph.D. student in Civil Engineering at the University of Maryland, College Park, advised by Prof. Yunfeng Zhang, working in computational solid mechanics. Builds constitutive models for metals, composites, and granular materials, calibrated against experimental data, and takes them through nonlinear finite element analysis in Abaqus and Ansys, with geometry modeled in SolidWorks. Applies machine learning, including deep neural networks, to predict material behavior from experimental and simulation data. First author of two peer-reviewed articles in ASCE's *International Journal of Geomechanics*, with additional co-authored work in the *ACI Materials Journal*.

EDUCATION

University of Maryland, College Park

Ph.D. in Civil Engineering

- Advisor: Prof. Yunfeng Zhang, Structural Engineering Laboratory

College Park, MD

2026 – Present

University of Connecticut

Ph.D. in Civil / Structural Engineering | GPA: 4.0/4.0

- Research: Constitutive modeling of composite and soft materials using finite element analysis and machine learning
- Coursework: Advanced Solid Mechanics, Polymer Properties, Finite Element I, Variable Topic Polymer Science

Storrs, CT

2024 – 2025

Dalhousie University

M.A.Sc. in Civil Engineering | GPA: 3.4/4.0

- Thesis: Numerical modeling of concrete-filled fiber-reinforced polymer piles
- Funded by a Natural Sciences and Engineering Research Council of Canada (NSERC) grant

Halifax, NS, Canada

2017 – 2020

Sharif University of Technology

M.Sc. in Civil / Structural Engineering | GPA: 16.28/20

- Thesis: Constitutive modeling of brittle materials (polymer concrete and rock) under triaxial compression loading
- Coursework: Linear & Nonlinear Finite Element Analysis, Design and Analysis of Composite Materials

Tehran, Iran

2014 – 2017

Iran University of Science and Technology

B.Sc. in Civil Engineering | GPA: 15.39/20

Tehran, Iran

2010 – 2014

RESEARCH INTERESTS

- Computational Solid Mechanics & Nonlinear Finite Element Analysis
- Constitutive Modeling of Metals, Composites, and Granular Materials
- Machine Learning and Deep Learning
- Data-Driven Materials Modeling
- Damage Initiation and Evolution in Fiber-Reinforced Composites
- Additive Manufacturing of Engineered Materials

EXPERIENCE

Doctoral Research Assistant

University of Maryland, College Park

- Characterized the mechanical and failure behavior of bamboo scrimber, a sustainable composite material, and calibrated anisotropic failure models in ANSYS for structural simulation of exoskeleton and humanoid robotic links.
- Reconstructed benchmark exoskeleton component geometry and evaluated bamboo scrimber against conventional aluminum alloys through comparative finite element structural analysis.
- Led a sponsored research project for the Maryland Department of Transportation on fatigue and failure of aluminum sign-post-clip connections, combining physical testing, 3D scanning, and finite element analysis to validate structural performance.

Jan. 2026 – Present

College Park, MD

Doctoral Research Assistant

University of Connecticut

- Performed finite element simulations of the elastic–plastic response of a randomized representative volume element (RVE) of a CFRP composite, including damage initiation and evolution.
- Built Python automation for FE geometry generation, meshing, and stochastic parameter randomization, allowing many composite configurations and material property variations to be analyzed in one batch.

Sep. 2024 – Dec. 2025

Storrs, CT

- Applied statistical sampling methods to the randomization of material properties, improving how well the simulated realizations span the property space.
- Surveyed CFRP durability and its constitutive modeling, and additive manufacturing / 3D printing techniques, to position the modeling work.

Data Analyst

Freelance

Nov. 2020 – Aug. 2024

Fredericton, NB, Canada

- Delivered data preprocessing and standardization projects, raising database accuracy and analytical reliability for client datasets.
- Analyzed large-scale user datasets and turned the results into actionable recommendations for client business decisions.

Graduate Research Assistant

Dalhousie University

Sep. 2017 – May 2020

Halifax, NS, Canada

- Modeled the axial behavior of FRP piles embedded in multi-layered soil in PLAXIS 3D.
- Modeled the lateral behavior of concrete-filled FRP tube piles with a MATLAB implementation of nonlinear finite element analysis, validated against full-scale field tests from a highway bridge on Route 40, Virginia; published in the *ASCE International Journal of Geomechanics*.
- Ran an experimental study on the effect of tire-derived aggregate on cylindrical concrete specimens under uniaxial compression.

Graduate Research Assistant

Sharif University of Technology

Sep. 2014 – Jan. 2017

Tehran, Iran

- Conducted experimental and numerical investigations of concrete behavior under uniaxial and triaxial compression, covering polymer concrete, ordinary cement concrete, lightweight concrete, and lime-mortar soil.
- Developed a disturbed state concept constitutive model for the softening response and volumetric deformation of rock under triaxial loading; published in the *ASCE International Journal of Geomechanics*.
- Contributed to peer review for journal submissions in materials engineering.

PUBLICATIONS

1. **Jafarian Abyaneh, M.**, El Naggar, H., and Sadeghian, P. (2020) "Numerical Modeling of the Lateral Behavior of Concrete-Filled FRP Tube Piles in Sand." *ASCE International Journal of Geomechanics*, Vol. 20, No. 8, 04020108. [doi:10.1061/\(ASCE\)GM.1943-5622.0001725](https://doi.org/10.1061/(ASCE)GM.1943-5622.0001725)
2. **Jafarian Abyaneh, M.** and Toufigh, V. (2018) "Softening Behavior and Volumetric Deformation of Rocks." *ASCE International Journal of Geomechanics*, Vol. 18, No. 8, 04018084. [doi:10.1061/\(ASCE\)GM.1943-5622.0001200](https://doi.org/10.1061/(ASCE)GM.1943-5622.0001200)
3. Toufigh, V., **Jafarian Abyaneh, M.**, and Jafari, K. (2017) "Study of Behavior of Concrete under Axial and Triaxial Compression." *ACI Materials Journal*, Vol. 114, No. 4, pp. 619–629. [doi:10.14359/51689716](https://doi.org/10.14359/51689716)
4. Jafari, K., **Jafarian Abyaneh, M.**, and Toufigh, V. (2017) "Experimental Study of Different Types of Concrete in Uniaxial Compression Test." *International Journal of Civil and Environmental Engineering*, Vol. 10, No. 12, pp. 1647–1651.
5. **Jafarian Abyaneh, M.**, Jafari, K., and Toufigh, V. (2016) "Constitutive Modeling of Different Types of Concrete under Uniaxial Compression." *International Journal of Civil and Environmental Engineering*, Vol. 10, No. 12, pp. 1483–1486.
6. **Jafarian Abyaneh, M.** (2019) "Numerical Modeling of Concrete-Filled Fiber-Reinforced Polymer Piles." Master's thesis, Dalhousie University, Halifax, NS, Canada.

CONFERENCE PAPERS

1. **Jafarian Abyaneh, M.**, El Naggar, H., and Sadeghian, P. (2018) "Soil–Structure Interaction Modeling of FRP Composite Piles." *Canadian Society for Civil Engineering (CSCE) Annual Conference*, Fredericton, NB, Canada.
2. **Jafarian Abyaneh, M.**, Sadeghian, P., and El Naggar, H. (2018) "Tire-Derived Aggregate Concrete for Bridge Applications." *10th International Conference on Short and Medium Span Bridges (SMSB)*, Canadian Society for Civil Engineering (CSCE), Quebec City, QC, Canada.
3. **Jafarian Abyaneh, M.**, El Naggar, H., and Sadeghian, P. (2017) "Lateral Behaviour of Concrete-Filled FRP Tube Piles." *GeoOttawa*, Ottawa, ON, Canada.

CERTIFICATIONS

- **Improving Deep Neural Networks** — DeepLearning.AI on Coursera (Apr. 2021)
- **Neural Networks and Deep Learning** — DeepLearning.AI on Coursera (Apr. 2021)
- **Machine Learning** — Stanford University on Coursera (Apr. 2021)

- **Python for Data Science and AI** — IBM on Coursera (Sep. 2020)

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, SQL, LaTeX, Typst, Git
- **Finite Element Packages:** Abaqus, Ansys Mechanical, PLAXIS 3D, SAP2000, ETABS
- **Computer-Aided Design:** SolidWorks, Ansys SpaceClaim, AutoCAD
- **Machine Learning & Data:** TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, Statistical Modeling